FLEXIBLE ROCKET AEROELASTIC PROGRAM (FLEXROC)

User's Guide

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TABLE OF CONTENTS

1 INTRODUCTION 1
2 INPUT FILE 3
3 PROGRAM EXECUTION 8
4 OUTPUT AND POSTPROCESSING 9
4.1 Output File 9
4.2 Postprocessing 10
5 EXAMPLE MODELS 11
5.1 Estes Mean Machine 11
5.2 D Super-Roc Altitude 14

Appendix A – Sample output listing

ACRONYMS

CG	Center of gravity
CP	Center of pressure
CPU	Central processing unit
LE	Leading edge
Pa	Pascal (Newtons per square meter)
PSI	Pounds per square inch
SR	Super-Roc
SRA	Super-Roc Altitude

1 INTRODUCTION

Super-Roc (SR) models can buckle or go unstable during flight if the vehicle is too flexible. An R&D report[1] presented by the author at NARAM-52 demonstrated that Super-Roc failure can be caused by aeroelastic divergence. A follow-up R&D report[2] by the author at NARAM-53 described a new method for predicting the failure velocity of Super-Roc models. The new method calculates the modal frequencies of a flexible rocket structure coupled to aerodynamic terms. This approach is similar to how aeroelastic flutter and divergence analysis is performed for aircraft and launch vehicles in the aerospace industry.

The new modal aeroelastic method was implemented in a computer program called the Flexible Rocket Aeroelastic Program (FlexRoc). The program is easy to use and can quickly and efficiently analyze various designs. The primary output from FlexRoc are modal frequencies across the velocity range covering the expected flight envelope of the rocket. As illustrated in Figure 1-1, aeroelastic failure occurs at the velocity when the primary frequency drops to zero. FlexRoc has been shown to have good agreement between predicted behavior and actual flight results for both successful and failed SR vehicles (see the NARAM-53 R&D report for more information).

This User's Guide describes how to operate the FlexRoc program. Steps include:

- Prepare the input file
- Run the program
- Review and postprocess the output results

[1] Flanigan, C., "Aeroelastic Analysis of Super-Roc Vehicles," NARAM-52 R&D report, August, 2010.

[2] Flanigan, C., "Modal Aeroelastic Analysis of Super-Roc Vehicles," NARAM-53 R&D report, July, 2011.

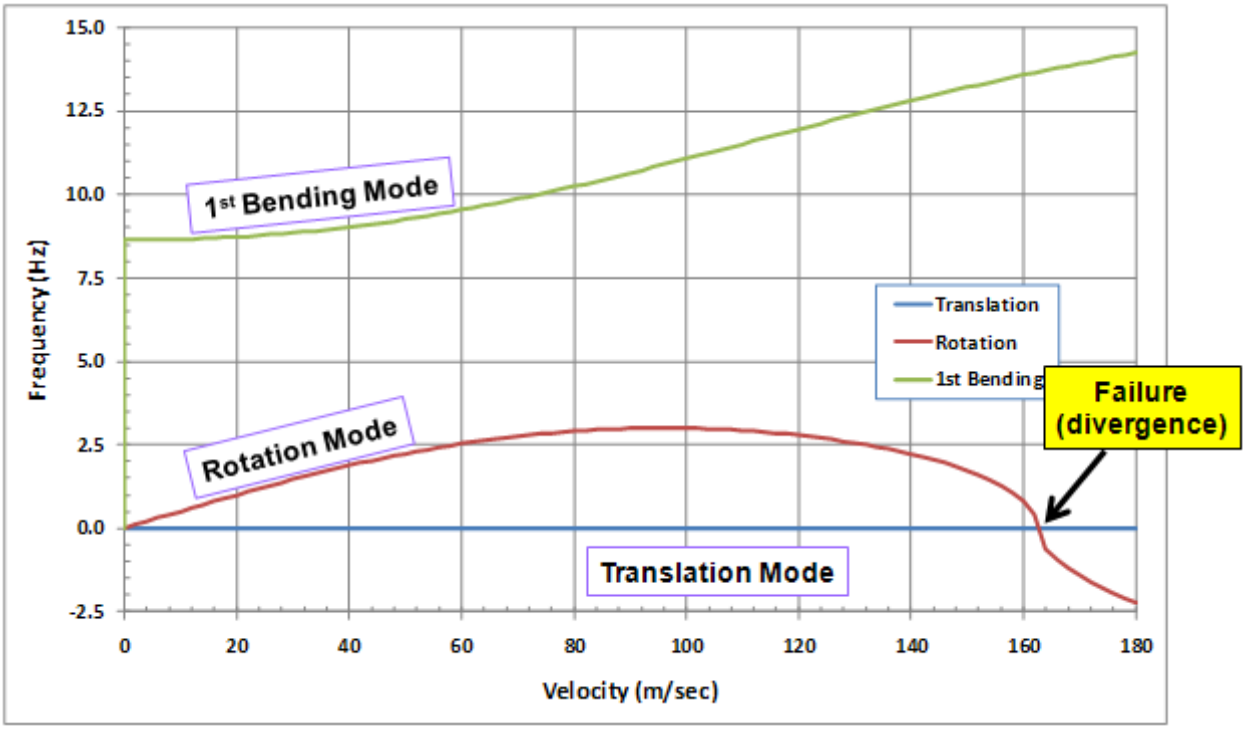

Figure 1-1. Aeroelastic failure occurs when the primary frequency drops to zero.

2 INPUT FILE

A FlexRoc input file defines a model rocket vehicle as illustrated in Figure 2-1. The model can include the following components:

- Nose cone (required)
- Tubes
- Transitions
- Mass objects (motor, parachutes, altimeters, etc.)
- Fins

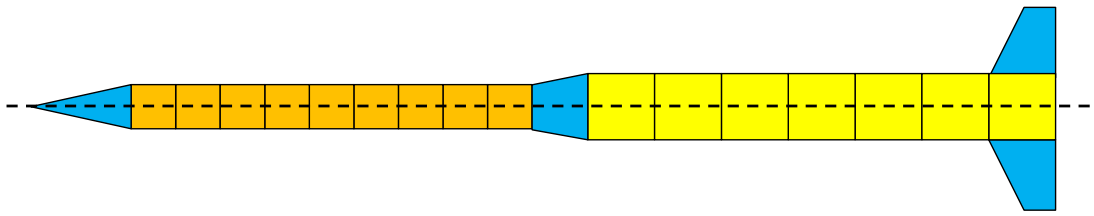

Figure 2-1. The FlexRoc input file defines a rocket vehicle from forward end (nose cone) to aft end (fins).

The FlexRoc input file is simple text file that can be created using a text editor (Notepad, EditPad, UltraEdit, etc.) or word processing program (Microsoft Word, etc.). If using a word processing program, be sure to save the file in text-only format.

FlexRoc input statements are listed in Table 2-1 and include the following:

- Comment (any line that starts with a $ in column 1).
- UNITS – defines the input units for following lines. Multiple UNITS statements are allowed.
- NOSE – defines the nose cone
- TUBE – defines a body tube
- TRAN – defines a conical transition
- MASS – defines a mass
- FINS – Defines trapezoidal fins (see Figure 2-2)
- VELOC – Defines the velocity range and launch altitude

Input statements use free field format with data entries separated by a comma.

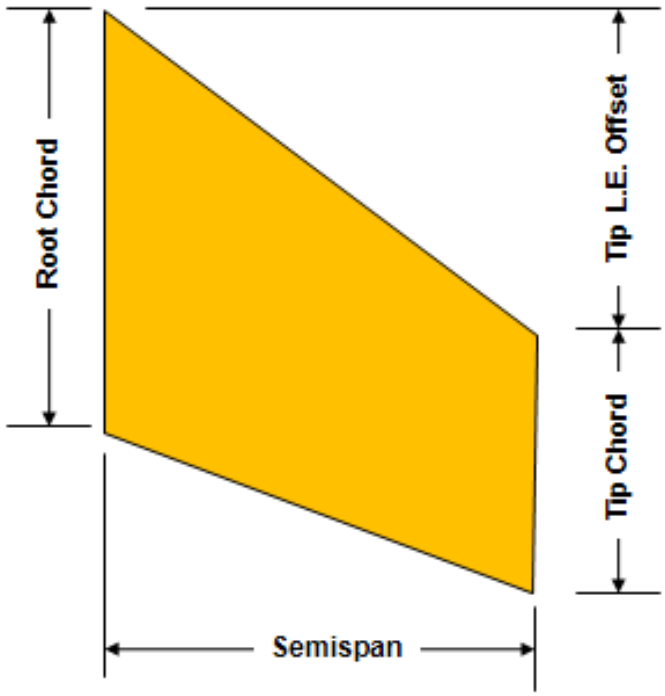

Figure 2-2. The fin planform is defined as a trapezoid.

A simple FlexRoc input deck is shown in Figure 2-3. The input file defines the model in sequential order starting from the front of the model (nose cone) to the end of the model (fins).

```
$  =======================
$  Excalibur B-C Super-Roc
$  =======================
$
$  Designed by Wolfram von Kiparski (NAR 28643)
$
$  Fwd section is 24" of T-5
$  Aft section is 24" of T-20
$
$  ============================================================
$
$  Units
$  -----
$
$   Length = inches
$   Mass   = grams
$   Force  = lbs
$
UNITS, inches, grams, lbs
$
$
$  Materials
$  ---------
$
$    Properties for paper tubes:
$    - Modulus = 217,557 PSI (1.5E9 Pa)
$    - Density = 16 grams/in^3
$
MATERIAL, paper, 217557, 16.0
$
$    Properties for 10 PCF balsa wood
$    - Modulus ~ 435,000 PSI (~3E9 Pa)
$    - Density = 10 PCF (2.6 grams/in^3)
$
MATERIAL, 10PCF_balsa, 435000, 2.6
$
$
$  Nose Cone
$  ---------
$
$   Length = 1.25"
$   O.D.   = 0.544"
$   Mass   = 0.5 gram
$
$  Length, OD, Mass
NOSE, 1.25, 0.544, 0.5
$
$
$  Body Tubes and Transition
$  -------------------------
$
$  T-5
$   Length = 24"
$   O.D.   = 0.544"
$   I.D.   = 0.518"
$
```

(continued)

Figure 2-3. Example input file.

```
$
$  Transition
$   Length = 0.75"
$
$  T-20
$   Length = 24"
$   O.D.   = 0.736"
$   I.D.   = 0.710"
$
$  Length, OD, ID, Material, NSM
TUBE, 24.0, 0.544, 0.518, paper, 0.0
$
$  Length, OD1, ID1, OD2, ID2, Material, NSM
TRAN, 0.75, 0.544, 0.001, 0.736, 0.001, 10PCF_Balsa, 0.0
$

$  Length, OD, ID, Material, NSM
TUBE, 24.0, 0.736, 0.710, paper, 0.0
$
$
$  Motor
$  -----
$
$   C6-7 = 24.2 grams
$
MASS, 24.2
$
$
$  Fins
$  ----
$
$   No. of fins   = 3
$   Root chord    = 1.75"
$   Tip chord     = 1.00"
$   Tip LE sweep  = 0.125"
$   Semispan      = 2.5"
$   Thickness     = 0.093"
$   Material      = Balsa
$
$  Nfin, Root Chord, Tip Chord, Tip LE sweep, span, thickness, material
FINS, 3, 1.75, 1.00, 0.125, 2.5, .093, 10PCF_Balsa
$
$
$  Solution
$  --------
$
$   Max veloc (C6-7) ~ 360 ft/sec
$   Analyze 0 to 600 ft/sec, increment = 2 ft/sec
$
$   Altitude = sea level
$
UNITS, ft, grams, lbs
$  V Start, V End, V Inc, Altitude
VELOC, 0.0, 600.0, 2.0, 0.0
$
```

Figure 2-3. (page 2 of 2)

Table 2.1. FlexRoc input statements.

Statement and Description
$ (comment)
Any line starting with a $ is a comment.
UNITS, length, mass, force
Defines the units used in subsequent statements. Options for length are **in**, **mm**, **cm**, and **m**. Options for mass are **oz**, **gr**, and **kg**. Options for force include **lb**, **pound**, and **Newton**. The values are converted from user units to SI units.
MATERIAL, name, E, density
Assigns a name for a material and defines the elastic modulus (E) and density of the material.
NOSE, length, OD, mass
Defines a nose cone. Input includes length, outside diameter, and mass.
TUBE, length, OD, ID, material, NSM
Defines a body tube. Input includes length, outside diameter, inside diameter, material, and nonstructural mass per unit length.
TRAN, length, OD1, ID1, OD2, ID2, material, NSM
Defines a conical transition. Input includes length, outside and inside diameter at the forward end, outside and inside diameter at the aft end, material, and nonstructural mass per unit length.
FINS, nfin, root chord, tip chord, tip LE offset, semispan, thick, material
Defines the number and shape of the fins. The fin shape is shown in Figure 2-2.
MASS, mass
Defines a lumped mass, such as a motor, altimeter, or parachute.
VELOC, V1, V2, inc, alt
Defines the velocity range and altitude at which solutions will be performed.

3 PROGRAM EXECUTION

FlexRoc is run as a simple "console" Windows application with an input file and an output file. To execute the program, do the following:

- Click on the "Start" menu and open a "Command Prompt" window.
- Use "cd" (change directory) commands to navigate to the folder where the FlexRoc executable file (flexroc.exe) and the input file are located.
- Execute FlexRoc by typing "flexroc" followed by hitting the "Enter" key. FlexRoc will execute and will prompt for the input file and output files.
- FlexRoc can also be executed by double-clicking on the file icon. You may need to enter the complete path name for the input and output files.

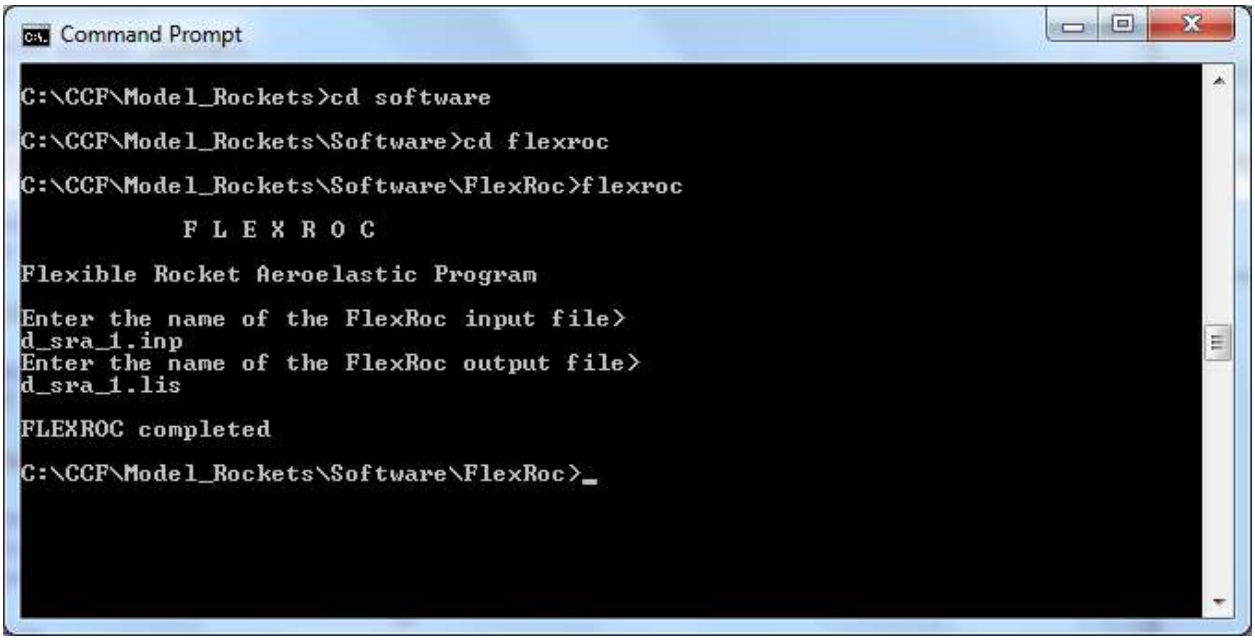

Figure 3-1. FlexRoc is executed within a Command Prompt window.

FlexRoc executes quickly. Run times are typically 10 seconds to one minute depending on the complexity of the model, the number of velocity solution points, and the CPU speed of your computer.

The output file produced by FlexRoc is a text file that can be easily viewed using a text editor or word processor. This is discussed in the next chapter.

4 OUTPUT AND POSTPROCESSING

4.1 OUTPUT FILE

The output file contains the following information:

- Echo of the input file
- Output of each input item (NOSE, TUBE, TRAN, etc.)
- Overall length and total vehicle mass (see Figure 4.1-1)
- Mass properties summary including the mass and CG station of each item, the total mass, and the station of the total CG (see Figure 4.1-1)
- Number of degrees of freedom in the structural finite element model (see Figure 4.1-1)
- Message for the solution being performed at each solution velocity (see Figure 4.1-1)
- A warning message is written if aeroelastic divergence occurs (see Figure 4.1-2)
- Solution summary listing the first three modal frequencies at each solution velocity (see Figure 4.1-3)

A complete output file is included in Appendix A.

```
Vehicle length = 1.57480E+00 m
Vehicle mass   = 8.31762E-02 kg

  M A S S   P R O P E R T I E S

Name       Mass (kg)   CG Sta (m)
-------- ------------ ------------
NOSE      1.00000E-03  1.69333E-02
TUBE      5.41001E-03  2.15900E-01
TRAN      2.08679E-04  4.12750E-01
TUBE      7.08668E-03  6.09600E-01
TRAN      2.97622E-04  8.06450E-01
TUBE      1.08913E-02  1.00330E+00
TUBE      1.08913E-02  1.38430E+00
MASS      4.50000E-02  1.57480E+00
FINS      2.39063E-03  1.59385E+00
         ------------ ------------
Total     8.31762E-02  1.28055E+00

Finite element model size =  70 degrees of freedom

Begin eigenvalue solutions

Solving at velocity =  0.00000E+00-- Dynamic pressure =  0.00000E+00
Solving at velocity =  7.62000E-01-- Dynamic pressure =  3.55644E-01
Solving at velocity =  1.52400E+00-- Dynamic pressure =  1.42258E+00
```

Figure 4.1-1. The FlexRoc output file includes vehicle length, vehicle mass, component mass, center of gravity location, and other information.

```
Solving at velocity =  9.98220E+01-- Dynamic pressure =  6.10321E+03
Solving at velocity =  1.00584E+02-- Dynamic pressure =  6.19675E+03
Solving at velocity =  1.01346E+02-- Dynamic pressure =  6.29099E+03
WARNING - Divergence likely occurred at velocity = 1.01346E+02
Solving at velocity =  1.02108E+02-- Dynamic pressure =  6.38595E+03
Solving at velocity =  1.02870E+02-- Dynamic pressure =  6.48162E+03
Solving at velocity =  1.03632E+02-- Dynamic pressure =  6.57800E+03
```

Figure 4.1-2. A warning message is written if the solution results indicate that aeroelastic divergence has occurred.

```
Solving at velocity =  1.50876E+02-- Dynamic pressure =  1.39427E+04
Solving at velocity =  1.51638E+02-- Dynamic pressure =  1.40839E+04
Solving at velocity =  1.52400E+02-- Dynamic pressure =  1.42258E+04

                 Summary of Results

  Velocity       Freq_1       Freq_2       Freq_3
   (m/sec)        (Hz)         (Hz)         (Hz)
 0.00000E+00 -5.66821E-04  1.13723E-03  9.36214E+00
 7.62000E-01  8.49660E-04  1.91215E-02  9.36211E+00
 1.52400E+00 -7.49080E-04  3.82678E-02  9.36202E+00
 2.28600E+00  8.93435E-04  5.73808E-02  9.36187E+00
 3.04800E+00 -2.06931E-04  7.65024E-02  9.36165E+00
 3.81000E+00  2.23191E-05  9.56057E-02  9.36138E+00
```

Figure 4.1-3. The results include the solution velocity and the frequencies of the first three modes.

4.2 POSTPROCESSING

FlexRoc results can be easily processed using spreadsheet or graphing programs such as Excel, OpenOffice, or similar. Copy the results summary from the FlexRoc output file to the spreadsheet. Use the spreadsheet program to prepare a summary graph as shown previously in Figure 1-1.

5 EXAMPLE MODELS

5.1 ESTES MEAN MACHINE

The Estes "Mean Machine" is a popular tall rocket. The model is "over 6 ft tall" as described on the kit packaging.

Design information for the Estes Mean Machine was obtained from a Rocksim file downloaded from EMRR RocketReviews.com[3]. Total length was 78.875". The mass of the airframe (w/o motor) was 7.14 oz.

Rocksim[4] was used to perform trajectory simulations. The maximum velocity using a D12 motor was 166.5 ft/sec (50.7 m/sec). The maximum velocity using an E9 motor was 221.8 ft/sec (67.6 m/sec).

The FlexRoc input file for the Estes Mean Machine is listed in Figure 5.1-1. The model consists of a nose cone followed by four tubes with couplers. The fins and the mass of the motor were defined at the end of the model.

The aeroelastic analysis results are shown in Figure 5.1-2. The predicted divergence speed was 143.9 m/sec. This provided very large margins between the expected maximum speed and the capability of the airframe using the standard motors. Successful flights of the Mean Machine have been reported using E30, F21, and F39 motors.

[3] http://www.rocketreviews.com/rocksim/estes_meanmachine.rkt
[4] Rocksim, Version: 9.1.0f114, Apogee Components, Inc. http://www.apogeerockets.com/rocksim.asp

```
$  ==================
$  Estes Mean Machine
$  ==================
$
$  Units
$  -----
$
$   Length = inches
$   Mass   = grams
$   Force  = lbs
$
UNITS, inches, grams, lbs
$
$
$  Materials
$  ---------
$
$    Properties for paper tubes:
$    - Modulus = 217,557 PSI (1.5E9 Pa)
$    - Density = 16 grams/in^3 (976.37 kg/m^3)
$
MATERIAL, paper, 217557, 16.0
$
$    Properties for 10 PCF balsa wood
$    - Modulus ~ 435,000 PSI (~3E9 Pa)
$    - Density = 10 PCF (2.6 grams/in^3)
$
MATERIAL, 10PCF_balsa, 435000, 2.6
$
$
$  Nose Cone
$  ---------
$
$  Length = 6.75"  (0.17145 m)
$  O.D.   = 1.637" (0.04158 m)
$  Mass   = 1.0 oz (28.35 gr)
$
$  Length, OD, Mass
NOSE, 6.75, 1.637, 28.35
$
$
$  Body Tubes
$  ----------
$
$  BT-60
$   Length = 18"    (0.45720 m)
$   O.D.   = 1.637" (0.04158 m)
$   I.D.   = 1.595" (0.04051 m)
$
$  Coupler = .1551 oz (4.4 gr)
$
```

(continued)

Figure 5.1-1. Input file for an Estes Mean Machine.

```
$  Length, OD, ID, Material, NSM
TUBE, 18.0, 1.637, 1.595, paper, 0.0
MASS, 4.4
TUBE, 18.0, 1.637, 1.595, paper, 0.0
MASS, 4.4
TUBE, 18.0, 1.637, 1.595, paper, 0.0
MASS, 4.4
TUBE, 18.0, 1.637, 1.595, paper, 0.0
MASS, 4.4
$
$
$  Motor
$  -----
$
$  Mass of E9-6 = 65 grams
$
MASS, 65.0
$
$
$  Fins
$  ----
$
$   Number of fins = 4
$   Root chord     = 2.625"
$   Tip chord      = 1.5"
$   Tip LE sweep   = 2.0"
$   Semi-span      = 1.875"  (0.04763 m)
$   Thickness      = 0.125"
$   Density        = 10 PCF
$
$  Nfin, Root Chord, Tip Chord, Tip LE sweep, span, thickness, material
FINS, 4, 2.625, 1.5, 2.0, 1.875, .125, 10PCF_balsa
$
$
$  Solution
$  --------
$
$   Max veloc (E9-6) = 221 ft/sec
$
$   Analyze 0 to 500 ft/sec, increment = 2 ft/sec
$
$   Altitude = sea level
$
UNITS, ft, grams, lbs
$  V Start, V End, V Inc, Altitude
VELOC, 0.0, 500.0, 2.0, 0.0
```

(continued)

Figure 5.1-1. (page 2 of 2).

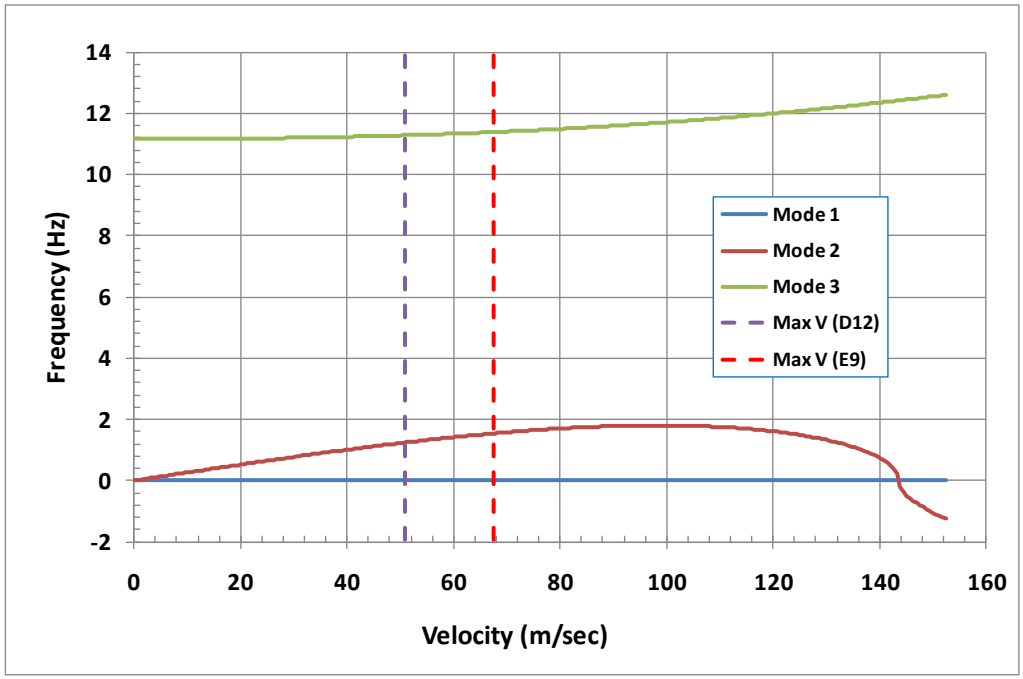

Figure 5.1-2. The Estes Mean Machine has a large margin between the aeroelastic limit and the expected maximum velocity.

5.2 D SUPER-ROC ALTITUDE

The modal aeroelastic method was used to help design a new model for D Super-Roc Altitude. The model was designed to disassemble into multiple sections to make it easier to transport and ship. The model also needed to carry a PerfectFlight Alt15K/WD altimeter for altitude measurement.

The model is shown in Figures 5.2-1 and 5.2-2. The model consisted of a T-50 aft section (two tubes that could be disassembled), a T-20 middle section, and a T-5 forward section. The total length of the model was 63.25" (160.7 cm). The weight of the airframe was 2.7 oz.

Rocksim predicted maximum altitude of 1,228 feet (374.3 m) and maximum velocity of 368.6 ft/sec (112.35 m/sec) assuming a "matt" surface finish for all parts.

The FlexRoc input file is shown in Figure 5.2-3. Note that an elastic modulus of paper was set to 250,000 PSI. This was based on data from testing of a TT-50 body tube, which was closer in size to the D super-roc model. The predicted aeroelastic performance is shown in Figure 5.2-4. The predicted divergence

velocity was 118.1 m/sec. This provided a small but positive margin of safety of +5.1%.

The model was flown for the first time at the contest held on 18-June-2011. The flight was successful with a very straight boost and no buckling. The model took first place with a score of 62,532 points. This increases confidence in the use of the modal aeroelastic method as a design tool for Super-Roc models.

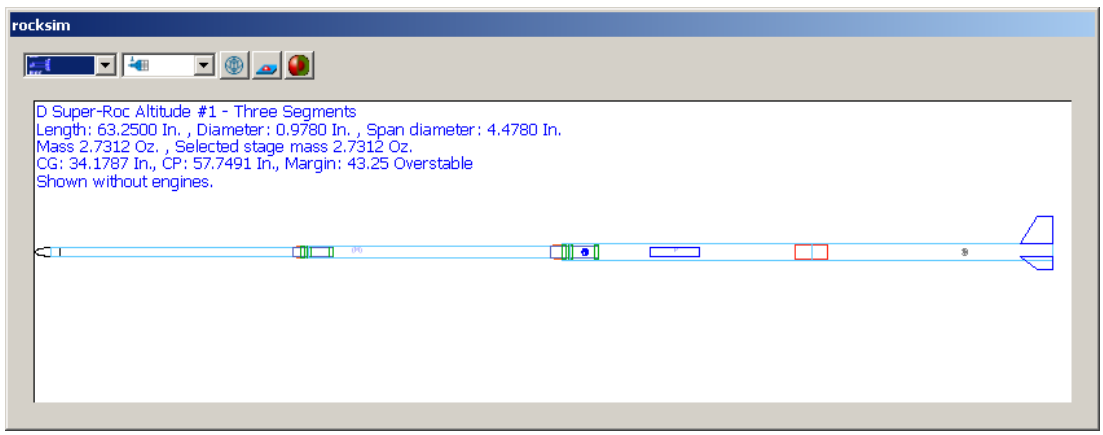

Figure 5.2-1. A new D Super-Roc Altitude model was assessed using the modal aeroelastic analysis method.

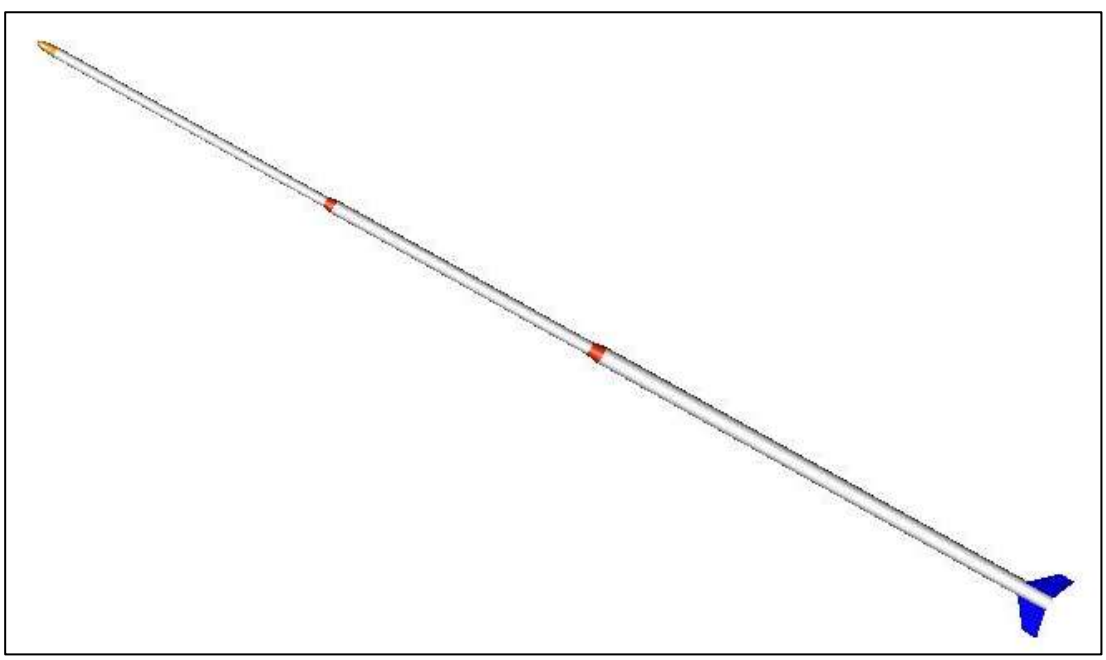

Figure 5.2-2. The model was designed to be disassembled for easy shipping.

```
$  ========================
$  D Super-Roc Altitude #1c
$  ========================
$
$  Same as D SRA #1b except with higher elastic modulus for paper
$
$  T-5  fwd section
$  T-20 mid section
$  T-50 aft sections
$
$==============================================================================
===
$
$
$  Units
$  -----
$
$   Length = inches
$   Mass   = grams
$   Force  = lbs
$
UNITS, inches, grams, lbs
$
$
$  Materials
$  ---------
$
$    Properties for paper tubes:
$    - Modulus = 250,000 PSI
$    - Density = 16 grams/in^3 (976.37 kg/m^3)
$
MATERIAL, paper, 250000, 16.0
$
$    Properties for 10 PCF balsa wood
$    - Modulus ~ 435,000 PSI (~3E9 Pa)
$    - Density = 10 PCF (2.6 grams/in^3)
$
MATERIAL, 10PCF_balsa, 435000, 2.6
$
$    Properties for basswood (for fins)
$    - Modulus ~ 1,000,000 PSI (~7E9 Pa)
$    - Density ~ 0.015 lbm/in^3 (6.8 grams/in^3)
$
MATERIAL, Basswood, 1000000, 6.8
$
$
$  Nose Cone
$  ---------
$
$   Length   = 1.0"
$   O.D.     = 0.545"
$   Mass     = 1 gram
$
$  Length, OD, Mass
NOSE, 1.0, 0.545, 1.0
```

(continued)

Figure 5.2-3. Input file for the D Super-Roc Altitude model.

```
$
$
$  Body Tubes
$  ----------
$
$  T-5
$   Length = 15"
$   O.D.   = 0.545"
$   I.D.   = 0.518"
$
$  Transition
$   Length = 0.5"
$
$  T-20
$   Length = 15"
$   O.D.   = 0.736"
$   I.D.   = 0.710"
$
$  Transition
$   Length = 0.75"
$
$  T-50
$   Length = 15"
$   O.D.   = 0.978"
$   I.D.   = 0.948"
$
$
$  Length, OD, ID, Material, NSM
TUBE, 15.0, 0.545, 0.518, paper, 0.0
TRAN,  0.5, 0.545, 0.518, 0.736, 0.710, paper, 0.0
TUBE, 15.0, 0.736, 0.710, paper, 0.0
TRAN,  0.5, 0.736, 0.710, 0.978, 0.948, paper, 0.0
TUBE, 18.0, 0.978, 0.948, paper, 0.0
TUBE, 14.0, 0.978, 0.948, paper, 0.0
$
$
$  Motor
$  -----
$
$   D12-7 = 45 grams
$
MASS, 45.0
$
$
$  Fins
$  ----
$
$   No. of fins   = 3
$   Root chord    = 2.0"
$   Tip chord     = 1.0"
$   Tip LE sweep  = 1.0"
$   Semispan      = 1.75"
$   Thickness     = 0.0625"
$   Density       = Basswood
```

(continued)

Figure 5.2-3. (page 2 of 3).

```
$
$  Nfin, Root Chord, Tip Chord, Tip LE sweep, span, thickness, material
FINS, 3, 2.0, 1.0, 1.0, 1.75, 0.0625, Basswood
$
$  Solution
$
$   Analyze 0 to 500 ft/sec, increment = 2.5 ft/sec
$
$   Altitude = sea level
$
UNITS, ft, grams, lbs
$  V Start, V End, V Inc, Altitude
VELOC, 0.0, 500.0, 2.5, 0.
$
```

Figure 5.2-3. (page 3 of 3).

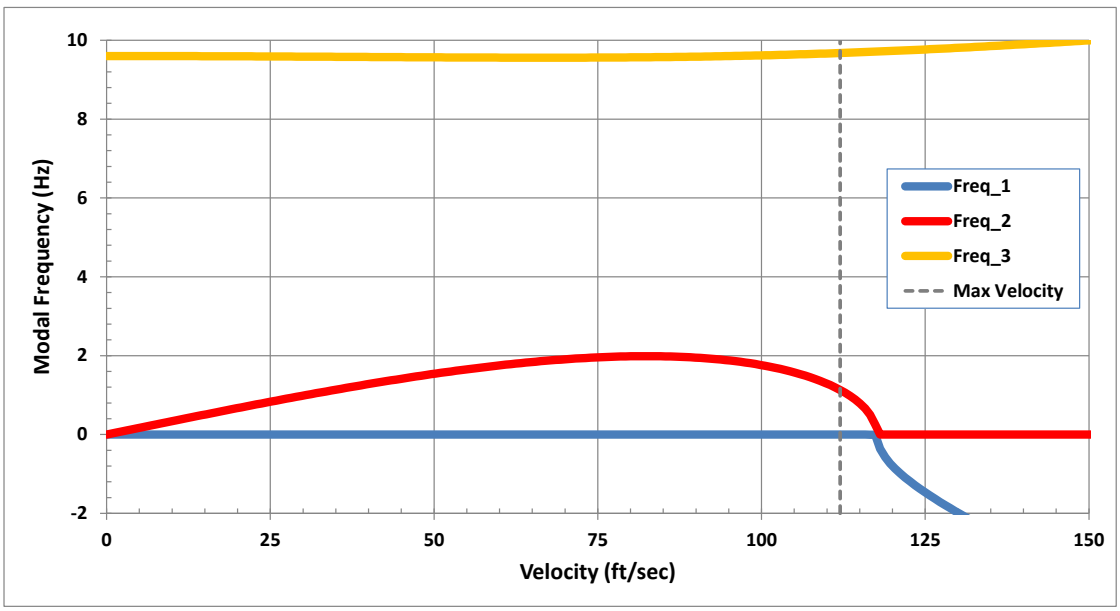

Figure 5.2-4. The D Super-Roc Altitude model was predicted to have a small positive margin of safety of 5.1%.

Appendix A

Sample Output Deck

Excalibur B/C Super-Roc

```
I N P U T   F I L E   E C H O
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$  =======================
$  Excalibur B-C Super-Roc
$  =======================
$
$  by Wolfram von Kiparski (NAR 28643)
$
$  Fwd section is 24" of T-5
$  Aft section is 24" of T-20
$
$  ============================================================
$
$  Units
$  -----
$
$   Length = inches
$   Mass   = grams
$   Force  = lbs
$
UNITS, inches, grams, lbs
$
$
$  Materials
$  ---------
$
$    Properties for paper tubes:
$    - Modulus = 217,557 PSI (1.5E9 Pa)
$    - Density = 16 grams/in^3 (976.37 kg/m^3)
$
MATERIAL, paper, 217557, 16.0
$
$    Properties for 10 PCF balsa wood
$    - Modulus ~ 435,000 PSI (~3E9 Pa)
$    - Density = 10 PCF (2.6 grams/in^3)
$
MATERIAL, 10PCF_balsa, 435000, 2.6
$
$
$  Nose Cone
$  ---------
$
$   Length = 1.25"
$   O.D.   = 0.544"
$   Mass   = 0.5 gram
$
$  Length, OD, Mass
NOSE, 1.25, 0.544, 0.5
$
$
$  Body Tubes, Transitions, and Boattail
$  -------------------------------------
$
$  T-5
$   Length = 24"
$   O.D.   = 0.544"
$   I.D.   = 0.518"
$
$  Transition
$   Length = 0.75"
$
$  T-20
$   Length = 24"
$   O.D.   = 0.736"
$   I.D.   = 0.710"
$
$  Length, OD, ID, Material, NSM
TUBE, 24.0, 0.544, 0.518, paper, 0.0
$
$  Length, OD1, ID1, OD2, ID2, Material, NSM
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TRAN, 0.75, 0.544, 0.001, 0.736, 0.001, 10PCF_Balsa, 0.0
$
$  Length, OD, ID, Material, NSM
TUBE, 24.0, 0.736, 0.710, paper, 0.0
$
$
$  Motor
$
$   C6-7 = 24.2 grams
$
MASS, 24.2
$
$  Fins
$
$   No. of fins   = 3
$   Root chord    = 1.75"
$   Tip chord     = 1.00"
$   Tip LE sweep  = 0.125"
$   Semispan      = 2.5"
$   Thickness     = 0.093"
$   Material      = Balsa
$
$  Nfin, Root Chord, Tip Chord, Tip LE sweep, span, thickness, material
FINS, 3, 1.75, 1.00, 0.125, 2.5, .093, 10PCF_Balsa
$
$  Solution
$
$   Max veloc (C6-7) = 361 ft/sec
$   Analyze 0 to 600 ft/sec, increment = 2 ft/sec
$
$   Altitude = sea level
$
UNITS, ft, grams, lbs
$  V Start, V End, V Inc, Altitude
VELOC, 0.0, 600.0, 2.0, 0.0

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E N D   O F   I N P U T   F I L E

  11 items read

UNITS
 Length units      = inches
 Length multiplier =  2.54000E-02
 Mass units        = grams
 Mass multiplier   =  1.00000E-03
 Force units       = lbs
 Force multiplier  =  4.44822E+00

MATERIAL
 Name        = paper
 Elastic mod = 2.17557E+05 lb/in^2
 Density     = 1.60000E+01 gr/in^3

MATERIAL
 Name        = 10PCF_balsa
 Elastic mod = 4.35000E+05 lb/in^2
 Density     = 2.60000E+00 gr/in^3

NOSE
 Length      = 1.25000E+00 in
 O.D.        = 5.44000E-01 in
 Mass        = 5.00000E-01 gr
 X-sect area = 2.32428E-01 in^2
 Surf area   = 1.09314E+00 in^2
 CN (unit Q) = 2.99906E-04

TUBE
 Length      = 2.40000E+01 in
 O.D.        = 5.44000E-01 in
 I.D.        = 5.18000E-01 in
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```
 Material    = paper
 NSM         = 0.00000E+00 gr/in
 X-sect area = 2.16864E-02 in^2
 Volume      = 5.20474E-01 in^3
 Bending I   = 7.64799E-04 in^4
 Surf area   = 4.10166E+01 in^2
 No. of elem = 8

TRAN
 Length      = 7.50000E-01 in
 O.D. Fwd    = 5.44000E-01 in
 I.D. Fwd    = 1.00000E-03 in
 O.D. Aft    = 7.36000E-01 in
 I.D. Aft    = 1.00000E-03 in
 Material    = 10PCF_Balsa
 NSM         = 0.00000E+00 gr/in
 X area Fwd  = 2.32427E-01 in^2
 X area Aft  = 4.25446E-01 in^2
 Volume      = 2.43083E-01 in^3
 Surf area   = 1.52027E+00 in^2
 Fwd seg I   = 6.02916E-03
 Fwd seg len = 3.75000E-01
 CN (unit Q) = 2.49057E-04
 Aft seg I   = 1.09983E-02
 Aft seg len = 3.75000E-01

TUBE
 Length      = 2.40000E+01 in
 O.D.        = 7.36000E-01 in
 I.D.        = 7.10000E-01 in
 Material    = paper
 NSM         = 0.00000E+00 gr/in
 X-sect area = 2.95278E-02 in^2
 Volume      = 7.08668E-01 in^3
 Bending I   = 1.93001E-03 in^4
 Surf area   = 5.54931E+01 in^2
 No. of elem = 8

MASS
 Mass        = 2.42000E+01 gr

FINS
 No. of fins =           3
 Root chord  = 1.75000E+00 in
 Tip chord   = 1.00000E+00 in
 Tip LE sweep= 1.25000E-01 in
 Semi-span   = 2.50000E+00 in
 Thickness   = 9.30000E-02 in
 Material    = 10PCF_Balsa
 Sweep angle =-1.43210E+00 deg
 Area (1 fin)= 3.43750E+00 in^2
 CN (unit Q) = 1.23566E-02

UNITS
 Length units      = ft
 Length multiplier =  3.04800E-01
 Mass units        = grams
 Mass multiplier   =  1.00000E-03
 Force units       = lbs
 Force multiplier  =  4.44822E+00

VELOC
 Veloc start = 0.00000E+00 ft/sec
 Veloc end   = 6.00000E+02 ft/sec
 Increment   = 2.00000E+00 ft/sec
 Launch alt  = 0.00000E+00 ft

Vehicle length = 1.27000E+00 m
Vehicle mass   = 4.74919E-02 kg

  M A S S   P R O P E R T I E S
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```
Name       Mass (kg)   CG Sta (m)
-------- ------------ ------------
NOSE      5.00000E-04  2.11667E-02
TUBE      8.32758E-03  3.36550E-01
TRAN      6.32017E-04  6.50875E-01
TUBE      1.13387E-02  9.65200E-01
MASS      2.42000E-02  1.27000E+00
FINS      2.49356E-03  1.29222E+00
         ------------ ------------
Total     4.74919E-02  1.01333E+00

Finite element model size =  38 degrees of freedom

Begin eigenvalue solutions

Solving at velocity =  0.00000E+00-- Dynamic pressure =  0.00000E+00
Solving at velocity =  6.09600E-01-- Dynamic pressure =  2.27612E-01
Solving at velocity =  1.21920E+00-- Dynamic pressure =  9.10450E-01
Solving at velocity =  1.82880E+00-- Dynamic pressure =  2.04851E+00
Solving at velocity =  2.43840E+00-- Dynamic pressure =  3.64180E+00
Solving at velocity =  3.04800E+00-- Dynamic pressure =  5.69031E+00
Solving at velocity =  3.65760E+00-- Dynamic pressure =  8.19405E+00
Solving at velocity =  4.26720E+00-- Dynamic pressure =  1.11530E+01
Solving at velocity =  4.87680E+00-- Dynamic pressure =  1.45672E+01
Solving at velocity =  5.48640E+00-- Dynamic pressure =  1.84366E+01
Solving at velocity =  6.09600E+00-- Dynamic pressure =  2.27612E+01
Solving at velocity =  6.70560E+00-- Dynamic pressure =  2.75411E+01
Solving at velocity =  7.31520E+00-- Dynamic pressure =  3.27762E+01
Solving at velocity =  7.92480E+00-- Dynamic pressure =  3.84665E+01
Solving at velocity =  8.53440E+00-- Dynamic pressure =  4.46120E+01
Solving at velocity =  9.14400E+00-- Dynamic pressure =  5.12128E+01
Solving at velocity =  9.75360E+00-- Dynamic pressure =  5.82688E+01
Solving at velocity =  1.03632E+01-- Dynamic pressure =  6.57800E+01
Solving at velocity =  1.09728E+01-- Dynamic pressure =  7.37464E+01
Solving at velocity =  1.15824E+01-- Dynamic pressure =  8.21681E+01
Solving at velocity =  1.21920E+01-- Dynamic pressure =  9.10450E+01
Solving at velocity =  1.28016E+01-- Dynamic pressure =  1.00377E+02
Solving at velocity =  1.34112E+01-- Dynamic pressure =  1.10164E+02
Solving at velocity =  1.40208E+01-- Dynamic pressure =  1.20407E+02
Solving at velocity =  1.46304E+01-- Dynamic pressure =  1.31105E+02
Solving at velocity =  1.52400E+01-- Dynamic pressure =  1.42258E+02
Solving at velocity =  1.58496E+01-- Dynamic pressure =  1.53866E+02
Solving at velocity =  1.64592E+01-- Dynamic pressure =  1.65929E+02
Solving at velocity =  1.70688E+01-- Dynamic pressure =  1.78448E+02
Solving at velocity =  1.76784E+01-- Dynamic pressure =  1.91422E+02
Solving at velocity =  1.82880E+01-- Dynamic pressure =  2.04851E+02
Solving at velocity =  1.88976E+01-- Dynamic pressure =  2.18736E+02
Solving at velocity =  1.95072E+01-- Dynamic pressure =  2.33075E+02
Solving at velocity =  2.01168E+01-- Dynamic pressure =  2.47870E+02
Solving at velocity =  2.07264E+01-- Dynamic pressure =  2.63120E+02
Solving at velocity =  2.13360E+01-- Dynamic pressure =  2.78825E+02
Solving at velocity =  2.19456E+01-- Dynamic pressure =  2.94986E+02
Solving at velocity =  2.25552E+01-- Dynamic pressure =  3.11601E+02
Solving at velocity =  2.31648E+01-- Dynamic pressure =  3.28672E+02
Solving at velocity =  2.37744E+01-- Dynamic pressure =  3.46199E+02
Solving at velocity =  2.43840E+01-- Dynamic pressure =  3.64180E+02
Solving at velocity =  2.49936E+01-- Dynamic pressure =  3.82617E+02
Solving at velocity =  2.56032E+01-- Dynamic pressure =  4.01508E+02
Solving at velocity =  2.62128E+01-- Dynamic pressure =  4.20855E+02
Solving at velocity =  2.68224E+01-- Dynamic pressure =  4.40658E+02
Solving at velocity =  2.74320E+01-- Dynamic pressure =  4.60915E+02
Solving at velocity =  2.80416E+01-- Dynamic pressure =  4.81628E+02
Solving at velocity =  2.86512E+01-- Dynamic pressure =  5.02796E+02
Solving at velocity =  2.92608E+01-- Dynamic pressure =  5.24419E+02
Solving at velocity =  2.98704E+01-- Dynamic pressure =  5.46498E+02
Solving at velocity =  3.04800E+01-- Dynamic pressure =  5.69031E+02
Solving at velocity =  3.10896E+01-- Dynamic pressure =  5.92020E+02
Solving at velocity =  3.16992E+01-- Dynamic pressure =  6.15464E+02
Solving at velocity =  3.23088E+01-- Dynamic pressure =  6.39363E+02
Solving at velocity =  3.29184E+01-- Dynamic pressure =  6.63718E+02
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Solving at velocity =  3.35280E+01-- Dynamic pressure =  6.88528E+02
Solving at velocity =  3.41376E+01-- Dynamic pressure =  7.13793E+02
Solving at velocity =  3.47472E+01-- Dynamic pressure =  7.39513E+02
Solving at velocity =  3.53568E+01-- Dynamic pressure =  7.65688E+02
Solving at velocity =  3.59664E+01-- Dynamic pressure =  7.92319E+02
Solving at velocity =  3.65760E+01-- Dynamic pressure =  8.19405E+02
Solving at velocity =  3.71856E+01-- Dynamic pressure =  8.46946E+02
Solving at velocity =  3.77952E+01-- Dynamic pressure =  8.74942E+02
Solving at velocity =  3.84048E+01-- Dynamic pressure =  9.03394E+02
Solving at velocity =  3.90144E+01-- Dynamic pressure =  9.32301E+02
Solving at velocity =  3.96240E+01-- Dynamic pressure =  9.61663E+02
Solving at velocity =  4.02336E+01-- Dynamic pressure =  9.91480E+02
Solving at velocity =  4.08432E+01-- Dynamic pressure =  1.02175E+03
Solving at velocity =  4.14528E+01-- Dynamic pressure =  1.05248E+03
Solving at velocity =  4.20624E+01-- Dynamic pressure =  1.08366E+03
Solving at velocity =  4.26720E+01-- Dynamic pressure =  1.11530E+03
Solving at velocity =  4.32816E+01-- Dynamic pressure =  1.14739E+03
Solving at velocity =  4.38912E+01-- Dynamic pressure =  1.17994E+03
Solving at velocity =  4.45008E+01-- Dynamic pressure =  1.21295E+03
Solving at velocity =  4.51104E+01-- Dynamic pressure =  1.24641E+03
Solving at velocity =  4.57200E+01-- Dynamic pressure =  1.28032E+03
Solving at velocity =  4.63296E+01-- Dynamic pressure =  1.31469E+03
Solving at velocity =  4.69392E+01-- Dynamic pressure =  1.34951E+03
Solving at velocity =  4.75488E+01-- Dynamic pressure =  1.38479E+03
Solving at velocity =  4.81584E+01-- Dynamic pressure =  1.42053E+03
Solving at velocity =  4.87680E+01-- Dynamic pressure =  1.45672E+03
Solving at velocity =  4.93776E+01-- Dynamic pressure =  1.49337E+03
Solving at velocity =  4.99872E+01-- Dynamic pressure =  1.53047E+03
Solving at velocity =  5.05968E+01-- Dynamic pressure =  1.56802E+03
Solving at velocity =  5.12064E+01-- Dynamic pressure =  1.60603E+03
Solving at velocity =  5.18160E+01-- Dynamic pressure =  1.64450E+03
Solving at velocity =  5.24256E+01-- Dynamic pressure =  1.68342E+03
Solving at velocity =  5.30352E+01-- Dynamic pressure =  1.72280E+03
Solving at velocity =  5.36448E+01-- Dynamic pressure =  1.76263E+03
Solving at velocity =  5.42544E+01-- Dynamic pressure =  1.80292E+03
Solving at velocity =  5.48640E+01-- Dynamic pressure =  1.84366E+03
Solving at velocity =  5.54736E+01-- Dynamic pressure =  1.88486E+03
Solving at velocity =  5.60832E+01-- Dynamic pressure =  1.92651E+03
Solving at velocity =  5.66928E+01-- Dynamic pressure =  1.96862E+03
Solving at velocity =  5.73024E+01-- Dynamic pressure =  2.01118E+03
Solving at velocity =  5.79120E+01-- Dynamic pressure =  2.05420E+03
Solving at velocity =  5.85216E+01-- Dynamic pressure =  2.09768E+03
Solving at velocity =  5.91312E+01-- Dynamic pressure =  2.14161E+03
Solving at velocity =  5.97408E+01-- Dynamic pressure =  2.18599E+03
Solving at velocity =  6.03504E+01-- Dynamic pressure =  2.23083E+03
Solving at velocity =  6.09600E+01-- Dynamic pressure =  2.27612E+03
Solving at velocity =  6.15696E+01-- Dynamic pressure =  2.32187E+03
Solving at velocity =  6.21792E+01-- Dynamic pressure =  2.36808E+03
Solving at velocity =  6.27888E+01-- Dynamic pressure =  2.41474E+03
Solving at velocity =  6.33984E+01-- Dynamic pressure =  2.46186E+03
Solving at velocity =  6.40080E+01-- Dynamic pressure =  2.50943E+03
Solving at velocity =  6.46176E+01-- Dynamic pressure =  2.55745E+03
Solving at velocity =  6.52272E+01-- Dynamic pressure =  2.60594E+03
Solving at velocity =  6.58368E+01-- Dynamic pressure =  2.65487E+03
Solving at velocity =  6.64464E+01-- Dynamic pressure =  2.70426E+03
Solving at velocity =  6.70560E+01-- Dynamic pressure =  2.75411E+03
Solving at velocity =  6.76656E+01-- Dynamic pressure =  2.80441E+03
Solving at velocity =  6.82752E+01-- Dynamic pressure =  2.85517E+03
Solving at velocity =  6.88848E+01-- Dynamic pressure =  2.90638E+03
Solving at velocity =  6.94944E+01-- Dynamic pressure =  2.95805E+03
Solving at velocity =  7.01040E+01-- Dynamic pressure =  3.01017E+03
Solving at velocity =  7.07136E+01-- Dynamic pressure =  3.06275E+03
Solving at velocity =  7.13232E+01-- Dynamic pressure =  3.11579E+03
Solving at velocity =  7.19328E+01-- Dynamic pressure =  3.16928E+03
Solving at velocity =  7.25424E+01-- Dynamic pressure =  3.22322E+03
Solving at velocity =  7.31520E+01-- Dynamic pressure =  3.27762E+03
Solving at velocity =  7.37616E+01-- Dynamic pressure =  3.33247E+03
Solving at velocity =  7.43712E+01-- Dynamic pressure =  3.38778E+03
Solving at velocity =  7.49808E+01-- Dynamic pressure =  3.44355E+03
Solving at velocity =  7.55904E+01-- Dynamic pressure =  3.49977E+03
Solving at velocity =  7.62000E+01-- Dynamic pressure =  3.55644E+03
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Solving at velocity =  7.68096E+01-- Dynamic pressure =  3.61358E+03
Solving at velocity =  7.74192E+01-- Dynamic pressure =  3.67116E+03
Solving at velocity =  7.80288E+01-- Dynamic pressure =  3.72920E+03
Solving at velocity =  7.86384E+01-- Dynamic pressure =  3.78770E+03
Solving at velocity =  7.92480E+01-- Dynamic pressure =  3.84665E+03
Solving at velocity =  7.98576E+01-- Dynamic pressure =  3.90606E+03
Solving at velocity =  8.04672E+01-- Dynamic pressure =  3.96592E+03
Solving at velocity =  8.10768E+01-- Dynamic pressure =  4.02624E+03
Solving at velocity =  8.16864E+01-- Dynamic pressure =  4.08701E+03
Solving at velocity =  8.22960E+01-- Dynamic pressure =  4.14824E+03
Solving at velocity =  8.29056E+01-- Dynamic pressure =  4.20992E+03
Solving at velocity =  8.35152E+01-- Dynamic pressure =  4.27206E+03
Solving at velocity =  8.41248E+01-- Dynamic pressure =  4.33465E+03
Solving at velocity =  8.47344E+01-- Dynamic pressure =  4.39770E+03
Solving at velocity =  8.53440E+01-- Dynamic pressure =  4.46120E+03
Solving at velocity =  8.59536E+01-- Dynamic pressure =  4.52516E+03
Solving at velocity =  8.65632E+01-- Dynamic pressure =  4.58958E+03
Solving at velocity =  8.71728E+01-- Dynamic pressure =  4.65445E+03
Solving at velocity =  8.77824E+01-- Dynamic pressure =  4.71977E+03
Solving at velocity =  8.83920E+01-- Dynamic pressure =  4.78555E+03
Solving at velocity =  8.90016E+01-- Dynamic pressure =  4.85179E+03
Solving at velocity =  8.96112E+01-- Dynamic pressure =  4.91848E+03
Solving at velocity =  9.02208E+01-- Dynamic pressure =  4.98562E+03
Solving at velocity =  9.08304E+01-- Dynamic pressure =  5.05322E+03
Solving at velocity =  9.14400E+01-- Dynamic pressure =  5.12128E+03
Solving at velocity =  9.20496E+01-- Dynamic pressure =  5.18979E+03
Solving at velocity =  9.26592E+01-- Dynamic pressure =  5.25876E+03
Solving at velocity =  9.32688E+01-- Dynamic pressure =  5.32818E+03
Solving at velocity =  9.38784E+01-- Dynamic pressure =  5.39806E+03
Solving at velocity =  9.44880E+01-- Dynamic pressure =  5.46839E+03
Solving at velocity =  9.50976E+01-- Dynamic pressure =  5.53918E+03
Solving at velocity =  9.57072E+01-- Dynamic pressure =  5.61042E+03
Solving at velocity =  9.63168E+01-- Dynamic pressure =  5.68212E+03
Solving at velocity =  9.69264E+01-- Dynamic pressure =  5.75427E+03
Solving at velocity =  9.75360E+01-- Dynamic pressure =  5.82688E+03
Solving at velocity =  9.81456E+01-- Dynamic pressure =  5.89994E+03
Solving at velocity =  9.87552E+01-- Dynamic pressure =  5.97346E+03
Solving at velocity =  9.93648E+01-- Dynamic pressure =  6.04744E+03
Solving at velocity =  9.99744E+01-- Dynamic pressure =  6.12186E+03
Solving at velocity =  1.00584E+02-- Dynamic pressure =  6.19675E+03
Solving at velocity =  1.01194E+02-- Dynamic pressure =  6.27209E+03
Solving at velocity =  1.01803E+02-- Dynamic pressure =  6.34788E+03
Solving at velocity =  1.02413E+02-- Dynamic pressure =  6.42413E+03
Solving at velocity =  1.03022E+02-- Dynamic pressure =  6.50084E+03
Solving at velocity =  1.03632E+02-- Dynamic pressure =  6.57800E+03
Solving at velocity =  1.04242E+02-- Dynamic pressure =  6.65562E+03
Solving at velocity =  1.04851E+02-- Dynamic pressure =  6.73369E+03
Solving at velocity =  1.05461E+02-- Dynamic pressure =  6.81221E+03
Solving at velocity =  1.06070E+02-- Dynamic pressure =  6.89119E+03
Solving at velocity =  1.06680E+02-- Dynamic pressure =  6.97063E+03
Solving at velocity =  1.07290E+02-- Dynamic pressure =  7.05052E+03
Solving at velocity =  1.07899E+02-- Dynamic pressure =  7.13087E+03
Solving at velocity =  1.08509E+02-- Dynamic pressure =  7.21167E+03
Solving at velocity =  1.09118E+02-- Dynamic pressure =  7.29293E+03
Solving at velocity =  1.09728E+02-- Dynamic pressure =  7.37464E+03
Solving at velocity =  1.10338E+02-- Dynamic pressure =  7.45681E+03
Solving at velocity =  1.10947E+02-- Dynamic pressure =  7.53944E+03
Solving at velocity =  1.11557E+02-- Dynamic pressure =  7.62251E+03
Solving at velocity =  1.12166E+02-- Dynamic pressure =  7.70605E+03
Solving at velocity =  1.12776E+02-- Dynamic pressure =  7.79004E+03
Solving at velocity =  1.13386E+02-- Dynamic pressure =  7.87448E+03
Solving at velocity =  1.13995E+02-- Dynamic pressure =  7.95938E+03
Solving at velocity =  1.14605E+02-- Dynamic pressure =  8.04473E+03
Solving at velocity =  1.15214E+02-- Dynamic pressure =  8.13054E+03
Solving at velocity =  1.15824E+02-- Dynamic pressure =  8.21681E+03
Solving at velocity =  1.16434E+02-- Dynamic pressure =  8.30353E+03
Solving at velocity =  1.17043E+02-- Dynamic pressure =  8.39071E+03
Solving at velocity =  1.17653E+02-- Dynamic pressure =  8.47834E+03
Solving at velocity =  1.18262E+02-- Dynamic pressure =  8.56642E+03
Solving at velocity =  1.18872E+02-- Dynamic pressure =  8.65496E+03
Solving at velocity =  1.19482E+02-- Dynamic pressure =  8.74396E+03
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WARNING - Divergence likely occurred at velocity = 1.19482E+02
Solving at velocity =  1.20091E+02-- Dynamic pressure =  8.83341E+03
Solving at velocity =  1.20701E+02-- Dynamic pressure =  8.92332E+03
Solving at velocity =  1.21310E+02-- Dynamic pressure =  9.01368E+03
Solving at velocity =  1.21920E+02-- Dynamic pressure =  9.10450E+03
Solving at velocity =  1.22530E+02-- Dynamic pressure =  9.19577E+03
Solving at velocity =  1.23139E+02-- Dynamic pressure =  9.28750E+03
Solving at velocity =  1.23749E+02-- Dynamic pressure =  9.37968E+03
Solving at velocity =  1.24358E+02-- Dynamic pressure =  9.47232E+03
Solving at velocity =  1.24968E+02-- Dynamic pressure =  9.56541E+03
Solving at velocity =  1.25578E+02-- Dynamic pressure =  9.65896E+03
Solving at velocity =  1.26187E+02-- Dynamic pressure =  9.75297E+03
Solving at velocity =  1.26797E+02-- Dynamic pressure =  9.84743E+03
Solving at velocity =  1.27406E+02-- Dynamic pressure =  9.94234E+03
Solving at velocity =  1.28016E+02-- Dynamic pressure =  1.00377E+04
Solving at velocity =  1.28626E+02-- Dynamic pressure =  1.01335E+04
Solving at velocity =  1.29235E+02-- Dynamic pressure =  1.02298E+04
Solving at velocity =  1.29845E+02-- Dynamic pressure =  1.03265E+04
Solving at velocity =  1.30454E+02-- Dynamic pressure =  1.04237E+04
Solving at velocity =  1.31064E+02-- Dynamic pressure =  1.05214E+04
Solving at velocity =  1.31674E+02-- Dynamic pressure =  1.06195E+04
Solving at velocity =  1.32283E+02-- Dynamic pressure =  1.07180E+04
Solving at velocity =  1.32893E+02-- Dynamic pressure =  1.08171E+04
Solving at velocity =  1.33502E+02-- Dynamic pressure =  1.09165E+04
Solving at velocity =  1.34112E+02-- Dynamic pressure =  1.10164E+04
Solving at velocity =  1.34722E+02-- Dynamic pressure =  1.11168E+04
Solving at velocity =  1.35331E+02-- Dynamic pressure =  1.12177E+04
Solving at velocity =  1.35941E+02-- Dynamic pressure =  1.13189E+04
Solving at velocity =  1.36550E+02-- Dynamic pressure =  1.14207E+04
Solving at velocity =  1.37160E+02-- Dynamic pressure =  1.15229E+04
Solving at velocity =  1.37770E+02-- Dynamic pressure =  1.16255E+04
Solving at velocity =  1.38379E+02-- Dynamic pressure =  1.17286E+04
Solving at velocity =  1.38989E+02-- Dynamic pressure =  1.18322E+04
Solving at velocity =  1.39598E+02-- Dynamic pressure =  1.19362E+04
Solving at velocity =  1.40208E+02-- Dynamic pressure =  1.20407E+04
Solving at velocity =  1.40818E+02-- Dynamic pressure =  1.21456E+04
Solving at velocity =  1.41427E+02-- Dynamic pressure =  1.22510E+04
Solving at velocity =  1.42037E+02-- Dynamic pressure =  1.23569E+04
Solving at velocity =  1.42646E+02-- Dynamic pressure =  1.24631E+04
Solving at velocity =  1.43256E+02-- Dynamic pressure =  1.25699E+04
Solving at velocity =  1.43866E+02-- Dynamic pressure =  1.26771E+04
Solving at velocity =  1.44475E+02-- Dynamic pressure =  1.27848E+04
Solving at velocity =  1.45085E+02-- Dynamic pressure =  1.28929E+04
Solving at velocity =  1.45694E+02-- Dynamic pressure =  1.30015E+04
Solving at velocity =  1.46304E+02-- Dynamic pressure =  1.31105E+04
Solving at velocity =  1.46914E+02-- Dynamic pressure =  1.32200E+04
Solving at velocity =  1.47523E+02-- Dynamic pressure =  1.33299E+04
Solving at velocity =  1.48133E+02-- Dynamic pressure =  1.34403E+04
Solving at velocity =  1.48742E+02-- Dynamic pressure =  1.35511E+04
Solving at velocity =  1.49352E+02-- Dynamic pressure =  1.36624E+04
Solving at velocity =  1.49962E+02-- Dynamic pressure =  1.37742E+04
Solving at velocity =  1.50571E+02-- Dynamic pressure =  1.38864E+04
Solving at velocity =  1.51181E+02-- Dynamic pressure =  1.39991E+04
Solving at velocity =  1.51790E+02-- Dynamic pressure =  1.41122E+04
Solving at velocity =  1.52400E+02-- Dynamic pressure =  1.42258E+04
Solving at velocity =  1.53010E+02-- Dynamic pressure =  1.43398E+04
Solving at velocity =  1.53619E+02-- Dynamic pressure =  1.44543E+04
Solving at velocity =  1.54229E+02-- Dynamic pressure =  1.45692E+04
Solving at velocity =  1.54838E+02-- Dynamic pressure =  1.46846E+04
Solving at velocity =  1.55448E+02-- Dynamic pressure =  1.48005E+04
Solving at velocity =  1.56058E+02-- Dynamic pressure =  1.49168E+04
Solving at velocity =  1.56667E+02-- Dynamic pressure =  1.50336E+04
Solving at velocity =  1.57277E+02-- Dynamic pressure =  1.51508E+04
Solving at velocity =  1.57886E+02-- Dynamic pressure =  1.52685E+04
Solving at velocity =  1.58496E+02-- Dynamic pressure =  1.53866E+04
Solving at velocity =  1.59106E+02-- Dynamic pressure =  1.55052E+04
Solving at velocity =  1.59715E+02-- Dynamic pressure =  1.56242E+04
Solving at velocity =  1.60325E+02-- Dynamic pressure =  1.57437E+04
Solving at velocity =  1.60934E+02-- Dynamic pressure =  1.58637E+04
Solving at velocity =  1.61544E+02-- Dynamic pressure =  1.59841E+04
Solving at velocity =  1.62154E+02-- Dynamic pressure =  1.61049E+04
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Solving at velocity =  1.62763E+02-- Dynamic pressure =  1.62263E+04
Solving at velocity =  1.63373E+02-- Dynamic pressure =  1.63480E+04
Solving at velocity =  1.63982E+02-- Dynamic pressure =  1.64703E+04
Solving at velocity =  1.64592E+02-- Dynamic pressure =  1.65929E+04
Solving at velocity =  1.65202E+02-- Dynamic pressure =  1.67161E+04
Solving at velocity =  1.65811E+02-- Dynamic pressure =  1.68397E+04
Solving at velocity =  1.66421E+02-- Dynamic pressure =  1.69637E+04
Solving at velocity =  1.67030E+02-- Dynamic pressure =  1.70882E+04
Solving at velocity =  1.67640E+02-- Dynamic pressure =  1.72132E+04
Solving at velocity =  1.68250E+02-- Dynamic pressure =  1.73386E+04
Solving at velocity =  1.68859E+02-- Dynamic pressure =  1.74645E+04
Solving at velocity =  1.69469E+02-- Dynamic pressure =  1.75908E+04
Solving at velocity =  1.70078E+02-- Dynamic pressure =  1.77176E+04
Solving at velocity =  1.70688E+02-- Dynamic pressure =  1.78448E+04
Solving at velocity =  1.71298E+02-- Dynamic pressure =  1.79725E+04
Solving at velocity =  1.71907E+02-- Dynamic pressure =  1.81007E+04
Solving at velocity =  1.72517E+02-- Dynamic pressure =  1.82293E+04
Solving at velocity =  1.73126E+02-- Dynamic pressure =  1.83583E+04
Solving at velocity =  1.73736E+02-- Dynamic pressure =  1.84878E+04
Solving at velocity =  1.74346E+02-- Dynamic pressure =  1.86178E+04
Solving at velocity =  1.74955E+02-- Dynamic pressure =  1.87482E+04
Solving at velocity =  1.75565E+02-- Dynamic pressure =  1.88791E+04
Solving at velocity =  1.76174E+02-- Dynamic pressure =  1.90104E+04
Solving at velocity =  1.76784E+02-- Dynamic pressure =  1.91422E+04
Solving at velocity =  1.77394E+02-- Dynamic pressure =  1.92745E+04
Solving at velocity =  1.78003E+02-- Dynamic pressure =  1.94071E+04
Solving at velocity =  1.78613E+02-- Dynamic pressure =  1.95403E+04
Solving at velocity =  1.79222E+02-- Dynamic pressure =  1.96739E+04
Solving at velocity =  1.79832E+02-- Dynamic pressure =  1.98080E+04
Solving at velocity =  1.80442E+02-- Dynamic pressure =  1.99425E+04
Solving at velocity =  1.81051E+02-- Dynamic pressure =  2.00775E+04
Solving at velocity =  1.81661E+02-- Dynamic pressure =  2.02129E+04
Solving at velocity =  1.82270E+02-- Dynamic pressure =  2.03488E+04
Solving at velocity =  1.82880E+02-- Dynamic pressure =  2.04851E+04

                 Summary of Results

  Velocity       Freq_1        Freq_2        Freq_3
   (m/sec)        (Hz)          (Hz)          (Hz)
 0.00000E+00 -1.22882E-03 -4.37714E-04  1.10525E+01
 6.09600E-01  3.86606E-04  4.87293E-02  1.10526E+01
 1.21920E+00 -3.45315E-04  9.74555E-02  1.10528E+01
 1.82880E+00  1.85563E-04  1.46167E-01  1.10532E+01
 2.43840E+00  1.17502E-04  1.94861E-01  1.10538E+01
 3.04800E+00  1.77846E-04  2.43529E-01  1.10545E+01
 3.65760E+00 -1.10045E-04  2.92166E-01  1.10553E+01
 4.26720E+00 -3.01125E-04  3.40765E-01  1.10564E+01
 4.87680E+00 -1.88609E-04  3.89321E-01  1.10576E+01
 5.48640E+00 -2.85098E-04  4.37826E-01  1.10589E+01
 6.09600E+00 -2.13361E-04  4.86275E-01  1.10604E+01
 6.70560E+00 -1.05000E-04  5.34661E-01  1.10621E+01
 7.31520E+00  2.45156E-04  5.82978E-01  1.10640E+01
 7.92480E+00  3.70904E-04  6.31220E-01  1.10660E+01
 8.53440E+00 -1.05715E-04  6.79381E-01  1.10681E+01
 9.14400E+00  9.12234E-05  7.27453E-01  1.10705E+01
 9.75360E+00 -4.68181E-04  7.75431E-01  1.10729E+01
 1.03632E+01 -5.28883E-04  8.23308E-01  1.10756E+01
 1.09728E+01 -5.24439E-04  8.71079E-01  1.10784E+01
 1.15824E+01 -7.65532E-04  9.18736E-01  1.10814E+01
 1.21920E+01 -8.93003E-04  9.66273E-01  1.10846E+01
 1.28016E+01 -2.42494E-04  1.01368E+00  1.10879E+01
 1.34112E+01 -3.18599E-04  1.06096E+00  1.10914E+01
 1.40208E+01 -3.29105E-04  1.10810E+00  1.10951E+01
 1.46304E+01 -3.36456E-04  1.15509E+00  1.10990E+01
 1.52400E+01 -3.80759E-04  1.20194E+00  1.11030E+01
 1.58496E+01 -2.26780E-04  1.24862E+00  1.11072E+01
 1.64592E+01 -8.43349E-05  1.29514E+00  1.11116E+01
 1.70688E+01 -1.23980E-04  1.34148E+00  1.11162E+01
 1.76784E+01 -4.43194E-04  1.38765E+00  1.11209E+01
 1.82880E+01 -2.71098E-04  1.43363E+00  1.11258E+01
 1.88976E+01 -2.59857E-04  1.47943E+00  1.11309E+01
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1.95072E+01	-2.49270E-04	1.52502E+00	1.11362E+01
2.01168E+01	-4.16931E-04	1.57041E+00	1.11417E+01
2.07264E+01	-4.06080E-04	1.61559E+00	1.11474E+01
2.13360E+01	-3.98332E-04	1.66055E+00	1.11533E+01
2.19456E+01	-5.11971E-04	1.70529E+00	1.11593E+01
2.25552E+01	-6.12466E-04	1.74980E+00	1.11656E+01
2.31648E+01	-2.91279E-04	1.79407E+00	1.11720E+01
2.37744E+01	-3.37516E-04	1.83809E+00	1.11787E+01
2.43840E+01	-3.54998E-04	1.88186E+00	1.11856E+01
2.49936E+01	1.35958E-04	1.92538E+00	1.11926E+01
2.56032E+01	-2.01842E-04	1.96863E+00	1.11999E+01
2.62128E+01	-4.36296E-04	2.01161E+00	1.12073E+01
2.68224E+01	-3.65669E-04	2.05431E+00	1.12150E+01
2.74320E+01	-3.61870E-04	2.09673E+00	1.12229E+01
2.80416E+01	-2.81591E-04	2.13885E+00	1.12310E+01
2.86512E+01	-2.28523E-04	2.18068E+00	1.12393E+01
2.92608E+01	9.77567E-05	2.22220E+00	1.12479E+01
2.98704E+01	-3.35605E-04	2.26341E+00	1.12566E+01
3.04800E+01	-4.44739E-04	2.30430E+00	1.12656E+01
3.10896E+01	-5.45344E-04	2.34486E+00	1.12748E+01
3.16992E+01	-6.05363E-04	2.38509E+00	1.12842E+01
3.23088E+01	-6.63365E-04	2.42499E+00	1.12939E+01
3.29184E+01	-4.95550E-04	2.46453E+00	1.13038E+01
3.35280E+01	-3.45892E-04	2.50373E+00	1.13139E+01
3.41376E+01	-1.25035E-04	2.54256E+00	1.13242E+01
3.47472E+01	-2.79014E-04	2.58103E+00	1.13348E+01
3.53568E+01	-2.45786E-04	2.61912E+00	1.13457E+01
3.59664E+01	-3.33549E-04	2.65684E+00	1.13567E+01
3.65760E+01	-1.29322E-04	2.69416E+00	1.13680E+01
3.71856E+01	-7.18968E-05	2.73110E+00	1.13796E+01
3.77952E+01	-2.12222E-04	2.76763E+00	1.13914E+01
3.84048E+01	-2.93081E-04	2.80376E+00	1.14035E+01
3.90144E+01	-2.21093E-04	2.83947E+00	1.14158E+01
3.96240E+01	1.31146E-04	2.87476E+00	1.14284E+01
4.02336E+01	-3.23673E-04	2.90963E+00	1.14412E+01
4.08432E+01	-4.26640E-04	2.94406E+00	1.14543E+01
4.14528E+01	-5.15769E-04	2.97806E+00	1.14676E+01
4.20624E+01	-4.14032E-04	3.01160E+00	1.14813E+01
4.26720E+01	-2.75908E-04	3.04470E+00	1.14951E+01
4.32816E+01	-4.39089E-04	3.07733E+00	1.15093E+01
4.38912E+01	-4.28055E-04	3.10950E+00	1.15237E+01
4.45008E+01	-5.07159E-04	3.14119E+00	1.15383E+01
4.51104E+01	-5.71162E-04	3.17241E+00	1.15533E+01
4.57200E+01	-6.08808E-04	3.20315E+00	1.15685E+01
4.63296E+01	-6.90177E-04	3.23339E+00	1.15840E+01
4.69392E+01	-2.97811E-04	3.26313E+00	1.15998E+01
4.75488E+01	-4.13037E-04	3.29238E+00	1.16158E+01
4.81584E+01	-1.85264E-04	3.32111E+00	1.16321E+01
4.87680E+01	-1.78690E-04	3.34933E+00	1.16487E+01
4.93776E+01	-1.22801E-04	3.37703E+00	1.16656E+01
4.99872E+01	-3.62207E-04	3.40420E+00	1.16828E+01
5.05968E+01	-2.98917E-04	3.43084E+00	1.17002E+01
5.12064E+01	-2.64177E-04	3.45694E+00	1.17179E+01
5.18160E+01	-2.62989E-04	3.48250E+00	1.17359E+01
5.24256E+01	-2.68836E-04	3.50750E+00	1.17542E+01
5.30352E+01	-2.78216E-04	3.53196E+00	1.17728E+01
5.36448E+01	-2.62258E-04	3.55585E+00	1.17916E+01
5.42544E+01	-2.69279E-04	3.57918E+00	1.18107E+01
5.48640E+01	-1.39703E-04	3.60193E+00	1.18301E+01
5.54736E+01	-2.68324E-04	3.62411E+00	1.18498E+01
5.60832E+01	-5.01367E-05	3.64571E+00	1.18698E+01
5.66928E+01	-2.49981E-04	3.66672E+00	1.18901E+01
5.73024E+01	-3.55804E-04	3.68715E+00	1.19106E+01
5.79120E+01	-9.36376E-05	3.70697E+00	1.19314E+01
5.85216E+01	-3.47280E-04	3.72620E+00	1.19525E+01
5.91312E+01	-2.13872E-04	3.74482E+00	1.19739E+01
5.97408E+01	1.49137E-04	3.76283E+00	1.19956E+01
6.03504E+01	-2.60782E-04	3.78022E+00	1.20175E+01
6.09600E+01	1.86650E-04	3.79699E+00	1.20397E+01
6.15696E+01	-3.09811E-04	3.81314E+00	1.20622E+01
6.21792E+01	-2.05651E-04	3.82866E+00	1.20850E+01

6.27888E+01	-6.00223E-05	3.84355E+00	1.21080E+01
6.33984E+01	-6.46064E-05	3.85780E+00	1.21313E+01
6.40080E+01	-3.39402E-04	3.87141E+00	1.21548E+01
6.46176E+01	-1.23240E-04	3.88437E+00	1.21787E+01
6.52272E+01	-1.71950E-04	3.89668E+00	1.22028E+01
6.58368E+01	-2.44300E-04	3.90834E+00	1.22271E+01
6.64464E+01	-2.26673E-04	3.91934E+00	1.22517E+01
6.70560E+01	-2.36569E-04	3.92967E+00	1.22766E+01
6.76656E+01	-2.28932E-04	3.93934E+00	1.23017E+01
6.82752E+01	-1.52274E-04	3.94833E+00	1.23270E+01
6.88848E+01	-2.65128E-04	3.95665E+00	1.23527E+01
6.94944E+01	-2.27669E-04	3.96429E+00	1.23785E+01
7.01040E+01	1.50860E-04	3.97125E+00	1.24046E+01
7.07136E+01	-2.79802E-04	3.97751E+00	1.24309E+01
7.13232E+01	-3.30269E-04	3.98309E+00	1.24575E+01
7.19328E+01	-6.01608E-05	3.98796E+00	1.24843E+01
7.25424E+01	-2.90007E-04	3.99213E+00	1.25113E+01
7.31520E+01	1.25153E-04	3.99560E+00	1.25385E+01
7.37616E+01	1.01032E-05	3.99836E+00	1.25660E+01
7.43712E+01	2.10327E-04	4.00040E+00	1.25937E+01
7.49808E+01	1.05928E-04	4.00172E+00	1.26215E+01
7.55904E+01	2.48903E-04	4.00231E+00	1.26496E+01
7.62000E+01	-1.86624E-04	4.00217E+00	1.26779E+01
7.68096E+01	-1.37681E-04	4.00130E+00	1.27064E+01
7.74192E+01	-2.14481E-04	3.99969E+00	1.27351E+01
7.80288E+01	-1.08043E-04	3.99733E+00	1.27640E+01
7.86384E+01	-2.25634E-04	3.99421E+00	1.27930E+01
7.92480E+01	7.88967E-05	3.99034E+00	1.28222E+01
7.98576E+01	-7.91011E-05	3.98570E+00	1.28517E+01
8.04672E+01	1.04370E-04	3.98030E+00	1.28812E+01
8.10768E+01	1.88049E-04	3.97411E+00	1.29110E+01
8.16864E+01	1.52030E-04	3.96714E+00	1.29409E+01
8.22960E+01	-3.11116E-04	3.95938E+00	1.29710E+01
8.29056E+01	1.93287E-04	3.95081E+00	1.30012E+01
8.35152E+01	-2.74366E-04	3.94144E+00	1.30315E+01
8.41248E+01	2.14012E-04	3.93125E+00	1.30620E+01
8.47344E+01	2.48257E-04	3.92024E+00	1.30927E+01
8.53440E+01	1.49648E-04	3.90839E+00	1.31235E+01
8.59536E+01	-1.56902E-04	3.89570E+00	1.31544E+01
8.65632E+01	-1.31237E-04	3.88215E+00	1.31854E+01
8.71728E+01	-1.48169E-04	3.86774E+00	1.32165E+01
8.77824E+01	5.04888E-05	3.85244E+00	1.32478E+01
8.83920E+01	3.24083E-04	3.83626E+00	1.32791E+01
8.90016E+01	3.22455E-04	3.81918E+00	1.33106E+01
8.96112E+01	-1.62932E-04	3.80118E+00	1.33421E+01
9.02208E+01	-1.62422E-04	3.78225E+00	1.33738E+01
9.08304E+01	9.33932E-05	3.76237E+00	1.34055E+01
9.14400E+01	-1.30441E-04	3.74153E+00	1.34373E+01
9.20496E+01	-2.47522E-04	3.71971E+00	1.34692E+01
9.26592E+01	1.36307E-04	3.69689E+00	1.35012E+01
9.32688E+01	-2.34819E-04	3.67306E+00	1.35332E+01
9.38784E+01	-2.28691E-04	3.64819E+00	1.35653E+01
9.44880E+01	-2.48745E-04	3.62225E+00	1.35974E+01
9.50976E+01	-3.43252E-04	3.59523E+00	1.36296E+01
9.57072E+01	-2.93356E-04	3.56711E+00	1.36619E+01
9.63168E+01	1.69997E-04	3.53784E+00	1.36942E+01
9.69264E+01	-2.45757E-04	3.50741E+00	1.37265E+01
9.75360E+01	-3.69879E-04	3.47579E+00	1.37588E+01
9.81456E+01	-2.81417E-04	3.44293E+00	1.37912E+01
9.87552E+01	-6.76955E-05	3.40881E+00	1.38236E+01
9.93648E+01	2.46325E-04	3.37338E+00	1.38560E+01
9.99744E+01	-2.52791E-04	3.33660E+00	1.38884E+01
1.00584E+02	-1.41151E-04	3.29843E+00	1.39209E+01
1.01194E+02	1.82152E-04	3.25881E+00	1.39533E+01
1.01803E+02	1.11398E-04	3.21770E+00	1.39857E+01
1.02413E+02	2.18811E-04	3.17504E+00	1.40181E+01
1.03022E+02	1.08992E-04	3.13075E+00	1.40506E+01
1.03632E+02	3.77186E-04	3.08478E+00	1.40830E+01
1.04242E+02	-2.95053E-04	3.03703E+00	1.41153E+01
1.04851E+02	2.49280E-04	2.98744E+00	1.41477E+01
1.05461E+02	2.91725E-04	2.93590E+00	1.41800E+01

1.06070E+02	2.18495E-04	2.88231E+00	1.42123E+01
1.06680E+02	3.35402E-04	2.82654E+00	1.42445E+01
1.07290E+02	3.58065E-04	2.76848E+00	1.42768E+01
1.07899E+02	-1.20656E-04	2.70796E+00	1.43089E+01
1.08509E+02	-2.03725E-04	2.64483E+00	1.43410E+01
1.09118E+02	-1.60901E-04	2.57889E+00	1.43731E+01
1.09728E+02	4.60050E-04	2.50990E+00	1.44051E+01
1.10338E+02	3.34819E-04	2.43763E+00	1.44370E+01
1.10947E+02	3.47994E-04	2.36175E+00	1.44689E+01
1.11557E+02	-1.22689E-04	2.28192E+00	1.45007E+01
1.12166E+02	7.38219E-04	2.19770E+00	1.45324E+01
1.12776E+02	5.53605E-04	2.10856E+00	1.45640E+01
1.13386E+02	6.39501E-04	2.01384E+00	1.45956E+01
1.13995E+02	5.02007E-04	1.91273E+00	1.46271E+01
1.14605E+02	7.86608E-04	1.80412E+00	1.46585E+01
1.15214E+02	7.54335E-04	1.68659E+00	1.46897E+01
1.15824E+02	9.12152E-04	1.55811E+00	1.47209E+01
1.16434E+02	1.37357E-03	1.41570E+00	1.47520E+01
1.17043E+02	1.10760E-03	1.25461E+00	1.47830E+01
1.17653E+02	1.47498E-03	1.06641E+00	1.48139E+01
1.18262E+02	3.26231E-03	8.32916E-01	1.48447E+01
1.18872E+02	6.30984E-03	4.93533E-01	1.48753E+01
1.19482E+02	-4.61704E-01	-6.74252E-03	1.49059E+01
1.20091E+02	-8.22641E-01	-2.45753E-03	1.49363E+01
1.20701E+02	-1.07097E+00	-2.92534E-03	1.49666E+01
1.21310E+02	-1.27434E+00	-6.91991E-04	1.49968E+01
1.21920E+02	-1.45177E+00	-9.22654E-04	1.50268E+01
1.22530E+02	-1.61186E+00	-7.02918E-05	1.50567E+01
1.23139E+02	-1.75935E+00	-9.79998E-04	1.50865E+01
1.23749E+02	-1.89719E+00	6.04116E-04	1.51161E+01
1.24358E+02	-2.02734E+00	-7.87920E-04	1.51456E+01
1.24968E+02	-2.15121E+00	-9.76176E-04	1.51750E+01
1.25578E+02	-2.26984E+00	9.74048E-04	1.52042E+01
1.26187E+02	-2.38400E+00	7.41625E-04	1.52332E+01
1.26797E+02	-2.49432E+00	1.00672E-03	1.52621E+01
1.27406E+02	-2.60128E+00	-4.13341E-04	1.52909E+01
1.28016E+02	-2.70529E+00	-6.36671E-04	1.53195E+01
1.28626E+02	-2.80668E+00	5.06373E-04	1.53479E+01
1.29235E+02	-2.90573E+00	6.46609E-04	1.53762E+01
1.29845E+02	-3.00267E+00	3.97806E-04	1.54043E+01
1.30454E+02	-3.09771E+00	-4.94269E-04	1.54322E+01
1.31064E+02	-3.19102E+00	-6.81887E-04	1.54600E+01
1.31674E+02	-3.28274E+00	-1.21455E-04	1.54876E+01
1.32283E+02	-3.37301E+00	6.48283E-04	1.55150E+01
1.32893E+02	-3.46196E+00	-3.49792E-04	1.55423E+01
1.33502E+02	-3.54967E+00	-2.46549E-04	1.55694E+01
1.34112E+02	-3.63625E+00	-5.80508E-04	1.55963E+01
1.34722E+02	-3.72177E+00	-3.64270E-04	1.56230E+01
1.35331E+02	-3.80632E+00	3.23632E-04	1.56495E+01
1.35941E+02	-3.88995E+00	-2.87265E-04	1.56759E+01
1.36550E+02	-3.97273E+00	2.15907E-04	1.57021E+01
1.37160E+02	-4.05471E+00	-5.85779E-04	1.57281E+01
1.37770E+02	-4.13595E+00	-6.96928E-04	1.57539E+01
1.38379E+02	-4.21650E+00	-5.61510E-04	1.57795E+01
1.38989E+02	-4.29638E+00	5.92128E-04	1.58049E+01
1.39598E+02	-4.37565E+00	4.24553E-04	1.58301E+01
1.40208E+02	-4.45434E+00	6.59695E-04	1.58551E+01
1.40818E+02	-4.53248E+00	5.59517E-04	1.58799E+01
1.41427E+02	-4.61010E+00	4.04249E-04	1.59046E+01
1.42037E+02	-4.68724E+00	2.24189E-04	1.59290E+01
1.42646E+02	-4.76391E+00	4.52045E-04	1.59532E+01
1.43256E+02	-4.84015E+00	-4.92040E-04	1.59773E+01
1.43866E+02	-4.91599E+00	-6.87158E-04	1.60011E+01
1.44475E+02	-4.99143E+00	4.42974E-04	1.60247E+01
1.45085E+02	-5.06650E+00	-9.55549E-04	1.60481E+01
1.45694E+02	-5.14122E+00	-6.30142E-04	1.60714E+01
1.46304E+02	-5.21561E+00	-6.85418E-04	1.60944E+01
1.46914E+02	-5.28969E+00	-6.81679E-04	1.61172E+01
1.47523E+02	-5.36347E+00	-5.71900E-04	1.61398E+01
1.48133E+02	-5.43697E+00	-6.96960E-04	1.61621E+01
1.48742E+02	-5.51020E+00	-4.89418E-04	1.61843E+01

1.49352E+02	-5.58318E+00	-4.16338E-04	1.62063E+01
1.49962E+02	-5.65592E+00	-2.94095E-04	1.62280E+01
1.50571E+02	-5.72843E+00	-4.25037E-04	1.62496E+01
1.51181E+02	-5.80072E+00	-2.74572E-04	1.62709E+01
1.51790E+02	-5.87281E+00	-4.92151E-04	1.62920E+01
1.52400E+02	-5.94471E+00	-4.28076E-04	1.63129E+01
1.53010E+02	-6.01642E+00	-3.76889E-04	1.63336E+01
1.53619E+02	-6.08795E+00	3.23650E-04	1.63540E+01
1.54229E+02	-6.15933E+00	2.66519E-04	1.63743E+01
1.54838E+02	-6.23054E+00	-2.39495E-04	1.63943E+01
1.55448E+02	-6.30161E+00	-5.95243E-04	1.64142E+01
1.56058E+02	-6.37254E+00	-2.71098E-04	1.64338E+01
1.56667E+02	-6.44334E+00	1.34649E-04	1.64531E+01
1.57277E+02	-6.51401E+00	-5.51378E-04	1.64723E+01
1.57886E+02	-6.58457E+00	-2.73848E-04	1.64913E+01
1.58496E+02	-6.65502E+00	-1.74280E-04	1.65100E+01
1.59106E+02	-6.72537E+00	-4.00357E-04	1.65285E+01
1.59715E+02	-6.79561E+00	-4.08485E-04	1.65468E+01
1.60325E+02	-6.86577E+00	-5.37499E-04	1.65649E+01
1.60934E+02	-6.93585E+00	-3.65561E-04	1.65828E+01
1.61544E+02	-7.00585E+00	-3.88078E-04	1.66004E+01
1.62154E+02	-7.07577E+00	-1.57120E-04	1.66179E+01
1.62763E+02	-7.14563E+00	-5.01128E-04	1.66351E+01
1.63373E+02	-7.21542E+00	-2.94283E-04	1.66521E+01
1.63982E+02	-7.28516E+00	-4.21723E-04	1.66689E+01
1.64592E+02	-7.35484E+00	-5.20987E-05	1.66854E+01
1.65202E+02	-7.42448E+00	-4.90841E-04	1.67018E+01
1.65811E+02	-7.49408E+00	2.07106E-04	1.67179E+01
1.66421E+02	-7.56363E+00	-4.32056E-04	1.67338E+01
1.67030E+02	-7.63315E+00	-3.42830E-04	1.67495E+01
1.67640E+02	-7.70264E+00	-5.49476E-04	1.67650E+01
1.68250E+02	-7.77211E+00	1.25807E-04	1.67803E+01
1.68859E+02	-7.84155E+00	-3.53383E-04	1.67954E+01
1.69469E+02	-7.91097E+00	-4.78867E-04	1.68102E+01
1.70078E+02	-7.98038E+00	2.46691E-04	1.68248E+01
1.70688E+02	-8.04977E+00	-3.13877E-04	1.68393E+01
1.71298E+02	-8.11916E+00	-6.18441E-04	1.68535E+01
1.71907E+02	-8.18855E+00	-4.61551E-04	1.68675E+01
1.72517E+02	-8.25793E+00	-3.46302E-04	1.68812E+01
1.73126E+02	-8.32731E+00	-6.96826E-04	1.68948E+01
1.73736E+02	-8.39670E+00	-4.95218E-04	1.69082E+01
1.74346E+02	-8.46610E+00	1.68147E-04	1.69213E+01
1.74955E+02	-8.53550E+00	-4.06097E-04	1.69343E+01
1.75565E+02	-8.60492E+00	-1.97800E-04	1.69470E+01
1.76174E+02	-8.67436E+00	-5.14159E-04	1.69596E+01
1.76784E+02	-8.74381E+00	-4.19986E-04	1.69719E+01
1.77394E+02	-8.81329E+00	-2.27236E-04	1.69840E+01
1.78003E+02	-8.88279E+00	-8.07392E-05	1.69959E+01
1.78613E+02	-8.95232E+00	1.58791E-04	1.70076E+01
1.79222E+02	-9.02187E+00	-3.48496E-04	1.70191E+01
1.79832E+02	-9.09146E+00	-1.91061E-04	1.70304E+01
1.80442E+02	-9.16108E+00	-3.00932E-04	1.70416E+01
1.81051E+02	-9.23074E+00	-4.60411E-04	1.70525E+01
1.81661E+02	-9.30044E+00	-2.30976E-04	1.70632E+01
1.82270E+02	-9.37017E+00	-5.59646E-04	1.70737E+01
1.82880E+02	-9.43995E+00	-1.65398E-04	1.70840E+01